Business Intelligence Systems

INFO 4555 / 5556

Fall 2025

Data Warehousing, Business Intelligence, and Dimensional Modeling Primer

Chapter 1



The goal of Business Intelligence Systems

Support Data Informed Decision Making

How does BI support Data Informed Decision Making?

- Consolidate disparate Data
- Organizing Data
- Cleaning Data
- Defining Data
- Providing historic reference for data
- Provide tools to slice and dice data (filter and aggregate data)
- Provide tools to visually represent data

BEFORE



AFTER



Organization Helps:

- How many shoes do I have?
- How many hats do I have?
- How many similar shirts?

Operational Systems
(OLTP) On Line Transaction Processing

- Designed to CAPTURE data
- Tend to process records one at a time
- Repetitive operations
- Process transactions
- Typically do not maintain history (current status only)
- Execute business processes
- Examples of Operational Systems: Finance, HR, Point of Sales, Inventory



BI / DW Systems
(OLAP) On Line Analytic Processing

- Designed to SHARE data
- Used to evaluate organization's performance
- Compare measures/counts over time (How are we doing compared to last month, year, etc?)
- Generally aggregate thousands, or more, records to get to final result set
- Generally preserve historical context versus only current state

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All of the following apply to Operational Systems, except:

A: Execute business processes

B: Designed for real time analytics

C: Process transactions

D: Tend to process records one at a time

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Database Tables (Entities)

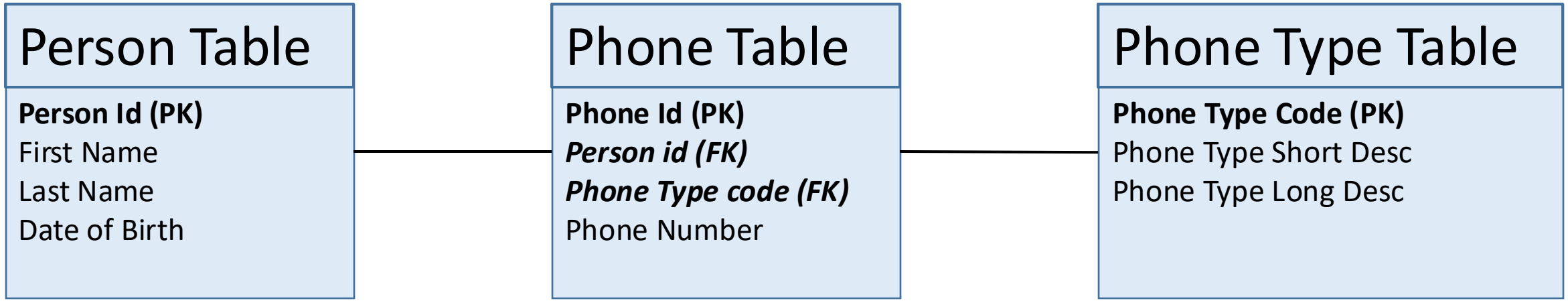
- A Table looks like a spreadsheet with rows and columns of data
- Each column in a table is an Attribute
- Each row in a table is Record
- Each Record should be uniquely identified by a Primary key

Sample Table

Person ID	First Name	Last Name	Date of Birth
01	Kaley	Cuoci	11/30/1985
02	Jim	Parson	03/24/1973
03	Johnny	Galecki	04/30/1975
04	Mayim	Bialik	12/12/1995
05	Kunal	Nayyar	04/30/1981
06	Simon	Helberg	12/09/1980



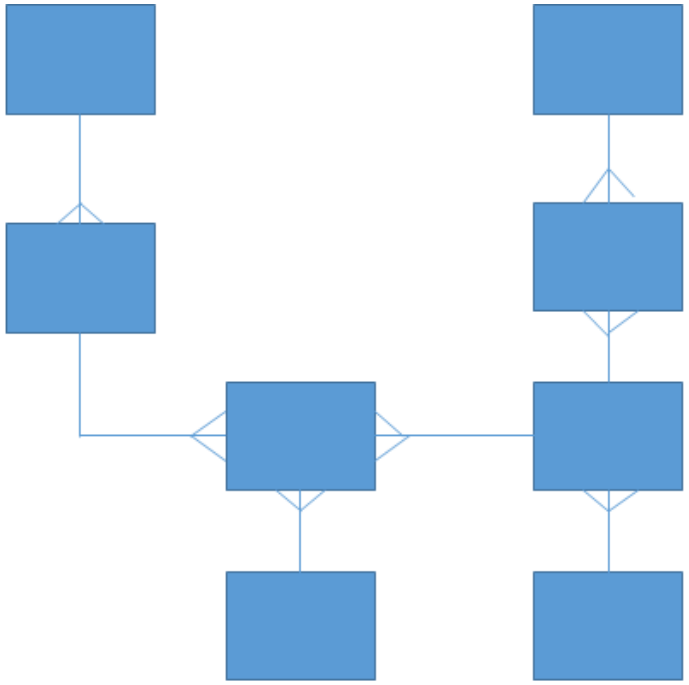
Sample Table Join



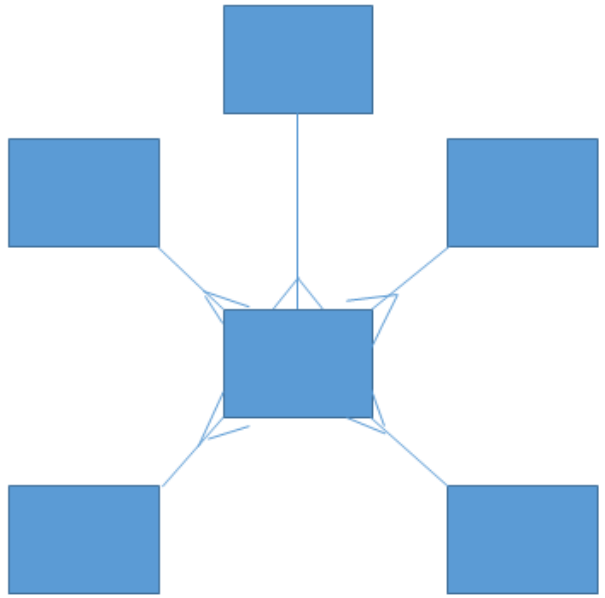
Sample SQL Table Join:

```
>SELECT *
  FROM PERSON_TABLE, PHONE_TABLE, PHONE_TYPE_TABLE
 WHERE PERSON_TABLE.PERSON_ID = PHONE_TABLE.PERSON_ID
 AND PHONE_TABLE.PHONE_TYPE_CODE= PHONE_TYPE_TABLE.PHONE_TYPE_CODE;
```

Operational 3NF Versus Dimensional Data Models

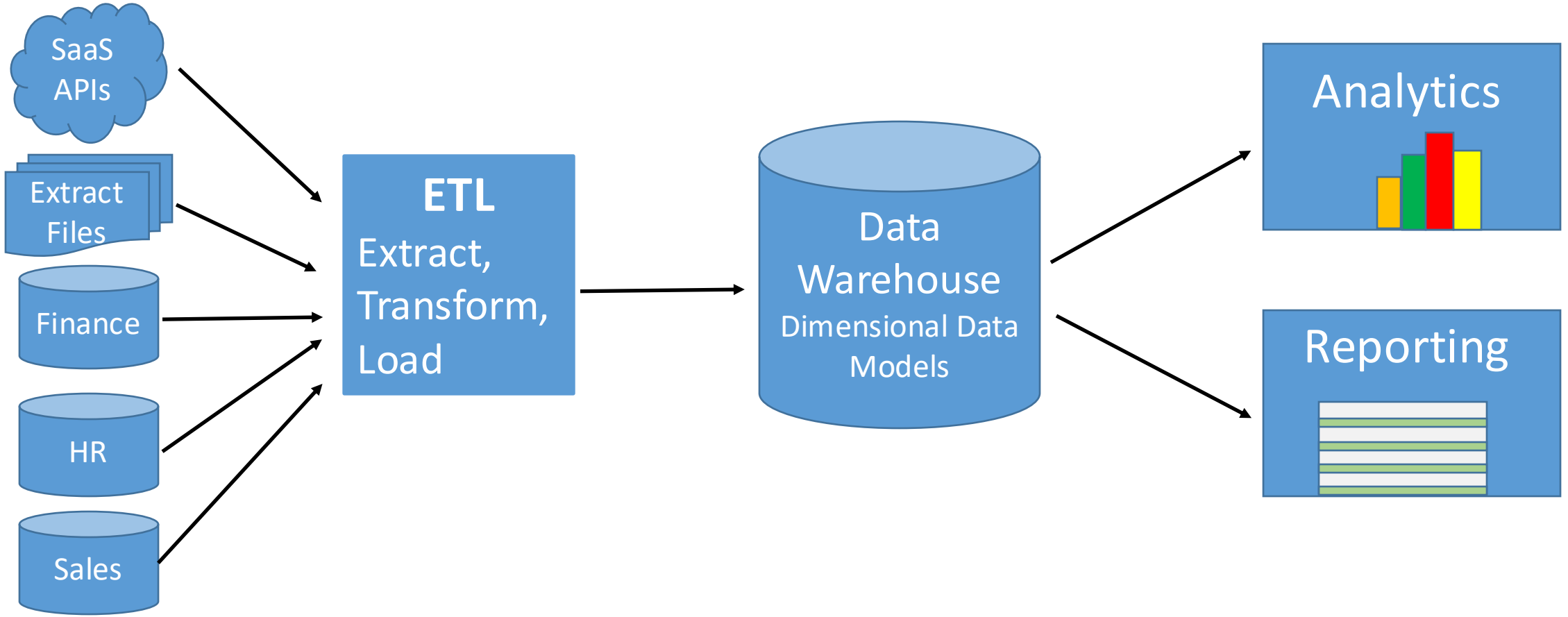


3NF “Normalized” Schema
Operational (transactional) OLTP



Dimensional (STAR) Schema
Data Warehouse OLAP

High Level Components of BI/DW System



A “read only” copy of the Operational Database is NOT a Data Warehouse!

- 3NF data models are not intuitive for Analytical type queries
- Operational databases typically do not maintain historic state, only current values
- Many values found in an operational system are not stored in the operation database, but derived via the application code.
- Other than taking the query load off the operational database, there is no value to this approach.



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A copy of the operational system database dedicated to reporting use is NOT a data warehouse for ALL of the following reasons except:

A: Table structures are not designed to simplify reporting and analytic queries

B: Historic state is not necessarily preserved

C: Is not designed to integrate with other data sources

D: Off loads reporting and analytics query workload from Operational database

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Goals of a DW/BI System

- Must make information easily accessible (simple and fast)
- Must present information consistently (cleansed, quality assured, consistent naming, etc)
- Must adapt to change
- Must present information in timely way
- Must protect information assets (proper access controls)
- Must serve as trustworthy source of information (decision support system)
- Must be accepted by the business community to be successful



FACT Tables

- store the performance measurements resulting from an organization's business process events
- contain multiple foreign keys, a primary key (also called a composite key) which is a subset of the foreign keys, and the measures
- deep in the number of rows, but narrow in number of columns

Grain

- level of detail for each record in a fact table
- It is critical that all rows in a fact table be at the same grain to avoid double counting

Fact Table attributes almost always numeric and ideally additive

- Additive facts allow you to sum up records no matter how you slice the fact table
- semi-additive facts (such as balances) cannot be summed across a time dimension
- non-additive facts (such as unit price) cannot be summed at all

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The following is true about a Fact table:

A: Has many columns and few records

B: Contains performance measures

C: serve as the primary source of query constraints, grouping, and report labels

D: is used for outrigging

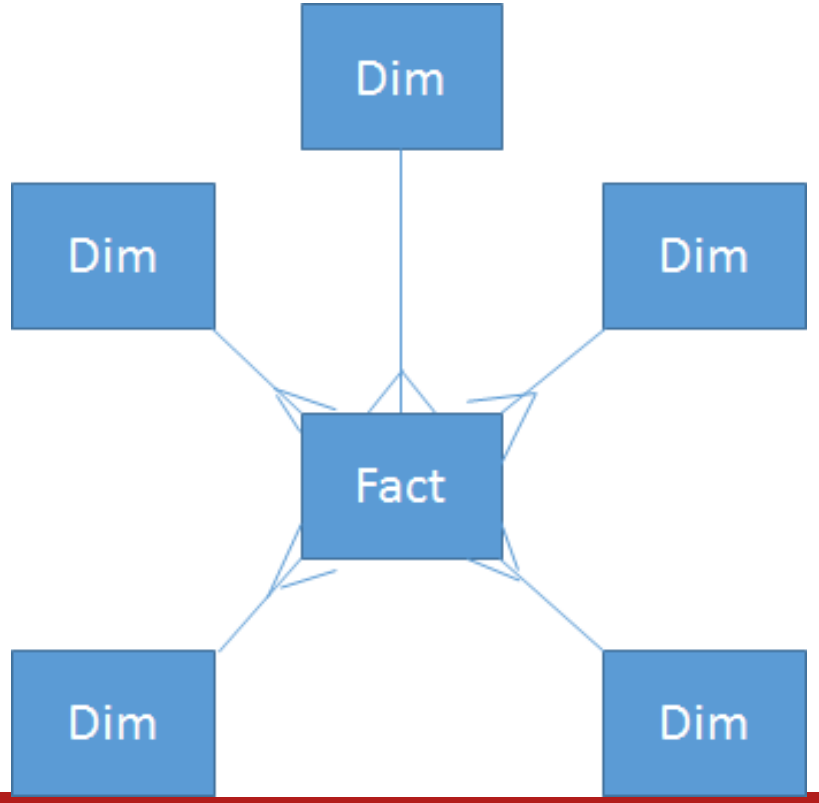
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Dimension Tables

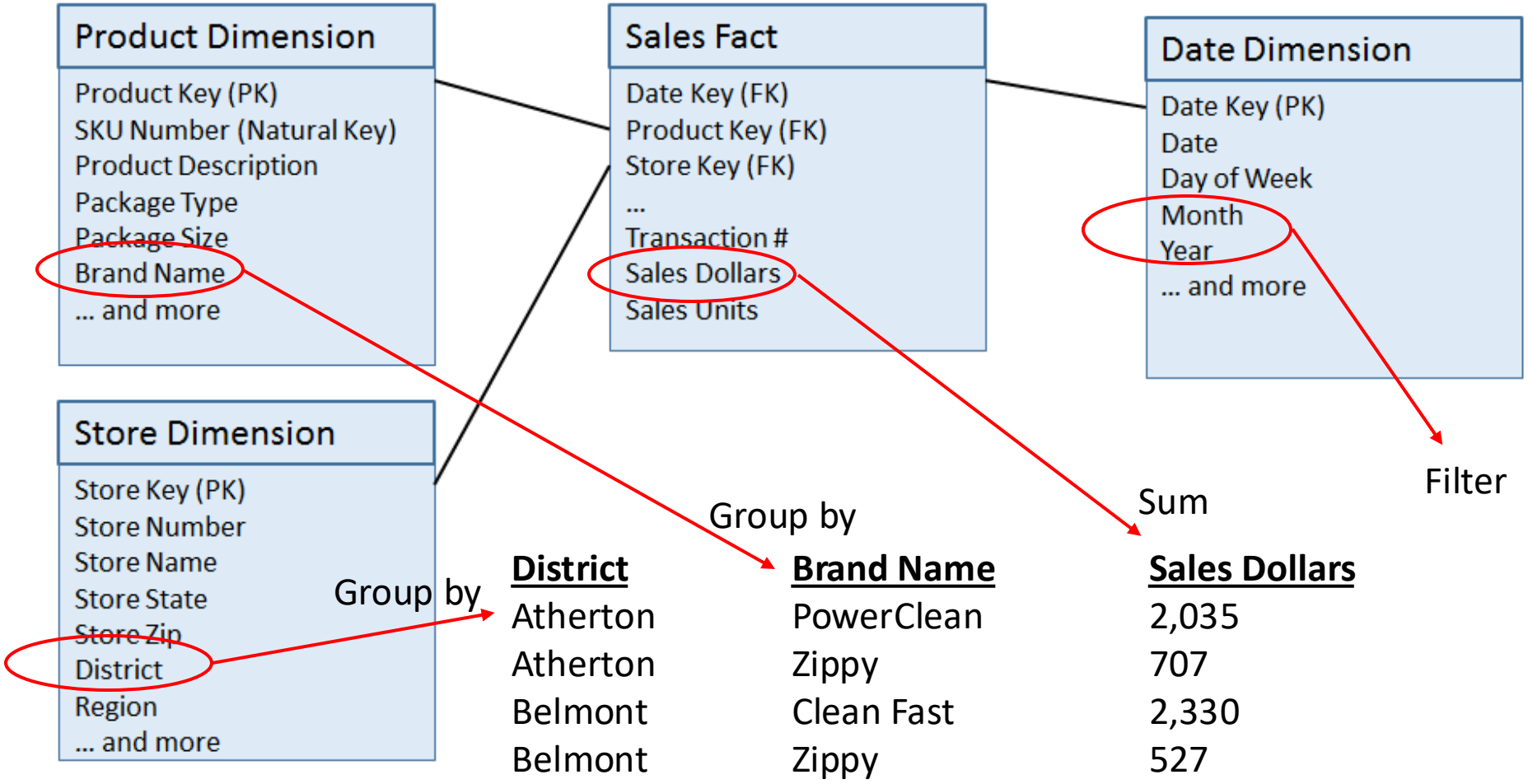
- contain textual context associated with the business process measurement event
- describe the who, what, when, where and why
- tend to have few rows, but are very wide with many textual columns
- Dimension attributes – serve as the primary source of query constraints, grouping, and report labels
- Dimension tables often contain hierarchies



Measurements go into fact tables, descriptive attributes into dimensions



Kimball Sales Model example



The Data Warehouse Toolkit, Ralph Kimball, Margy Ross – figure 1-6

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How comfortable are you feeling with Facts and Dimensions?

A: Got it! Let's keep going.

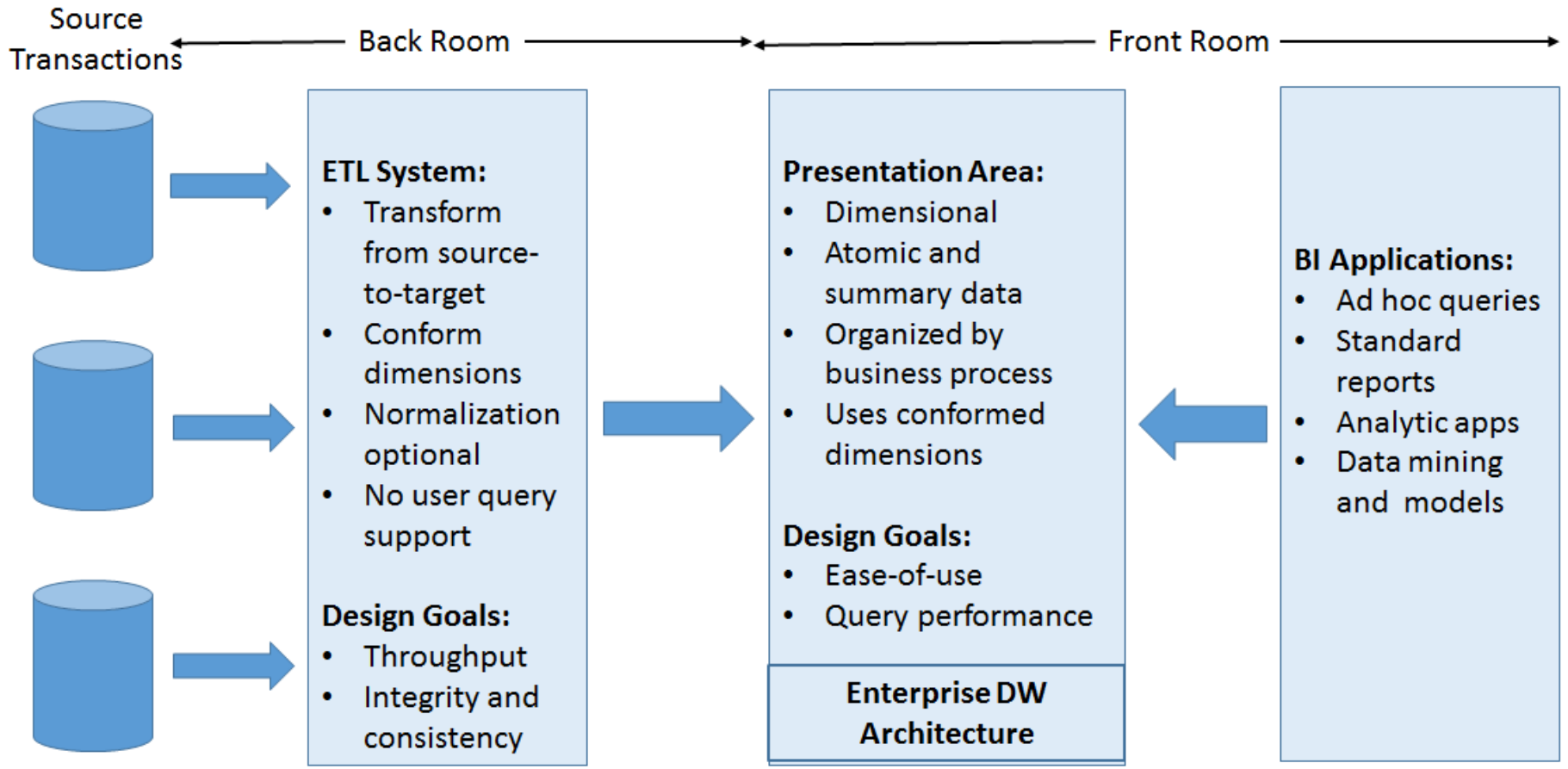
B: No Problem

C: It kind of makes sense

D: SLOW DOWN!

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Kimball DW/BI Architecture



The Data Warehouse Toolkit, Ralph Kimball, Margy Ross – figure 1-7

ETL (Extract, Transform, Load)

- The processing between the source operational system and the “front room” dimensional presentation tables.
- **Extract** – pulling the data from the source system to the Data Warehouse area
- **Transform** – cleansing data, deal with missing elements, parsing into standard formats, combining data from multiple sources, de-duplicating data
- **Load** – loading the transformed data into the final dimensional table structures

DW Presentation Area

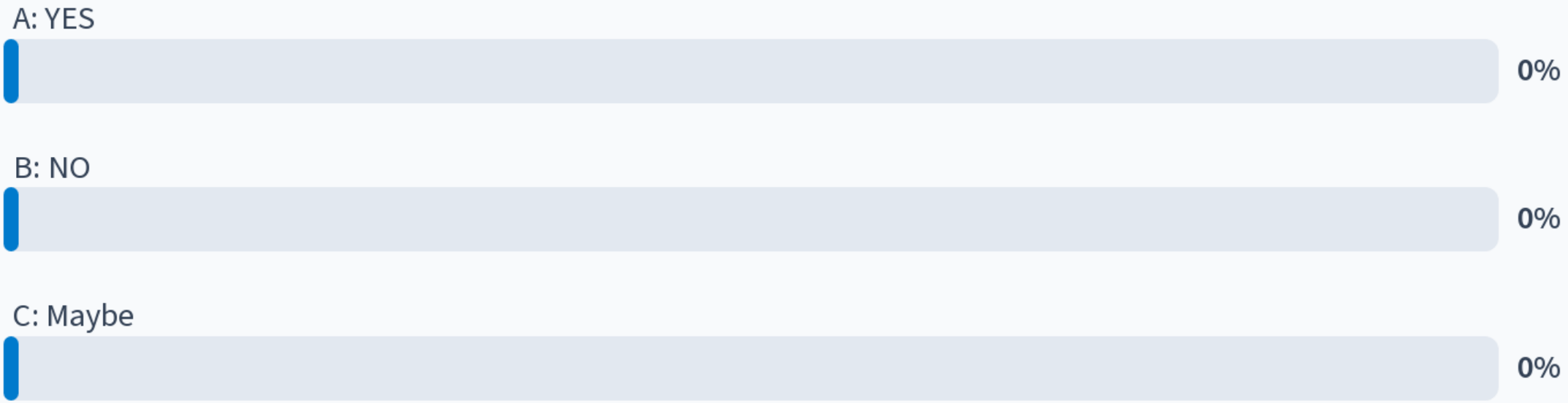
- Only the final dimensional tables available for end user queries
- at the atomic level
- business process centric
- adhere to the enterprise data warehouse bus architecture

Business Intelligence (BI) Applications

- used to query the presentation area
- Tools range from ad hoc, to pre-built parameter-driven applications that don’t allow users to construct queries directly

I feel that I am ready for the Concept Quiz. (Quiz on Tuesday)

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Don’t Forget…

- Buy your Textbook
- If you haven’t setup Poll Everywhere, please do so.
- No discussion/lab sections this Friday
- First quiz will be next Tuesday (bring laptop)